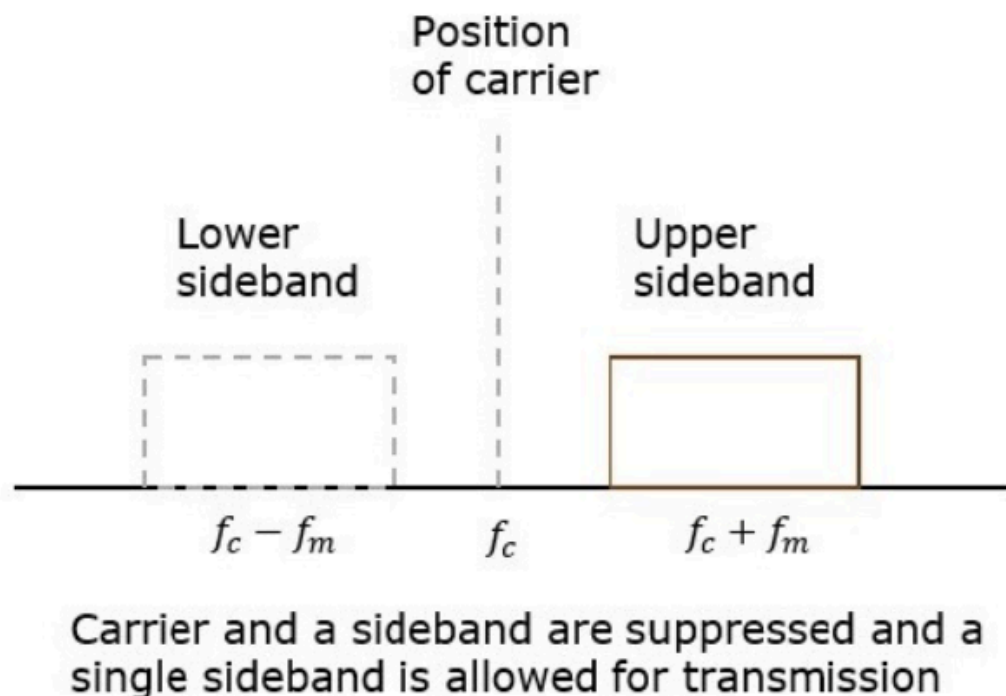


PRINCIPLES OF COMMUNICATION

PROJECT REPORT

Introduction:

Conceptually, the generation of a SSB signal is straightforward. We first generate a double-sideband signal and then apply an ideal band-pass filter to the result with cutoff frequencies of f_c and $f_c + W$ for the upper sideband, for instance. Practically, the approximate construction of an ideal filter is very difficult. The process of suppressing one of the sidebands along with the carrier and transmitting a single sideband is called a Single Sideband Suppressed Carrier system or simply SSBSC.



In the above figure, the carrier and the lower sideband are suppressed. Hence, the upper sideband is used for transmission. Similarly, we can suppress the carrier and the upper sideband while transmitting the lower sideband.

This SSB SC system, which transmits a single sideband has high power, as the power allotted for both the carrier and the other sideband is utilized in transmitting this Single Sideband.

Mathematical Expressions For SSB SC Signal:

Let us consider the mathematical expressions for the modulating and the carrier signals

Modulating signal

$$m(t) = A_m \cos(2\pi f_m t)$$

Carrier signal

$$c(t) = A_c \cos(2\pi f_c t)$$

Mathematically, we can represent the equation of SSB SC wave as

$$s(t) = \frac{A_m A_c}{2} \cos[2\pi(f_c + f_m)t] \text{ for the upper sideband}$$

Or

$$s(t) = \frac{A_m A_c}{2} \cos[2\pi(f_c - f_m)t] \text{ for the lower sideband}$$

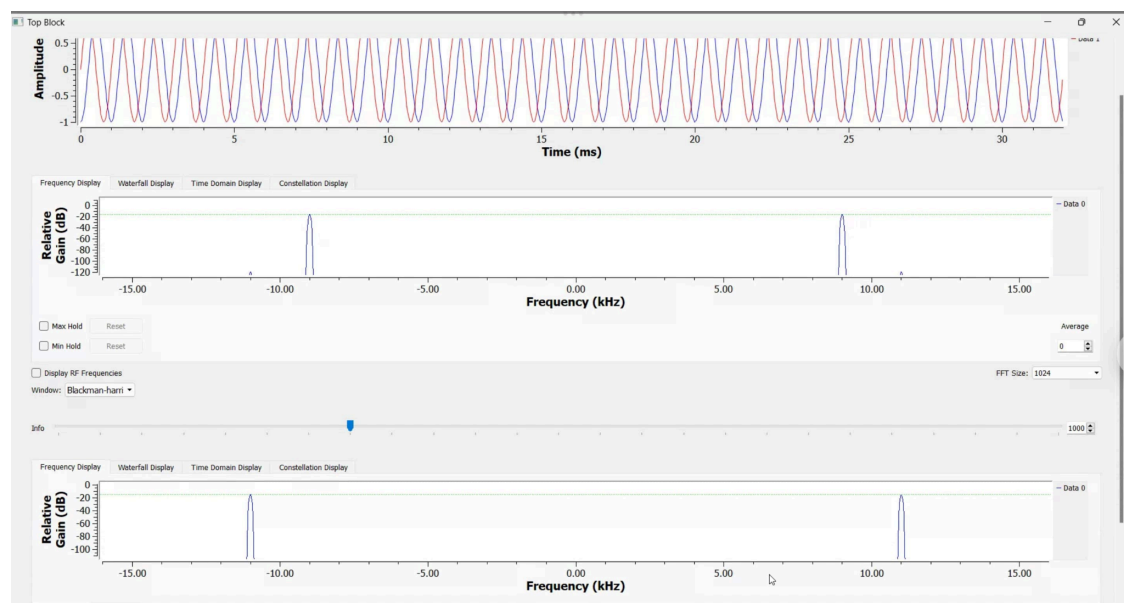
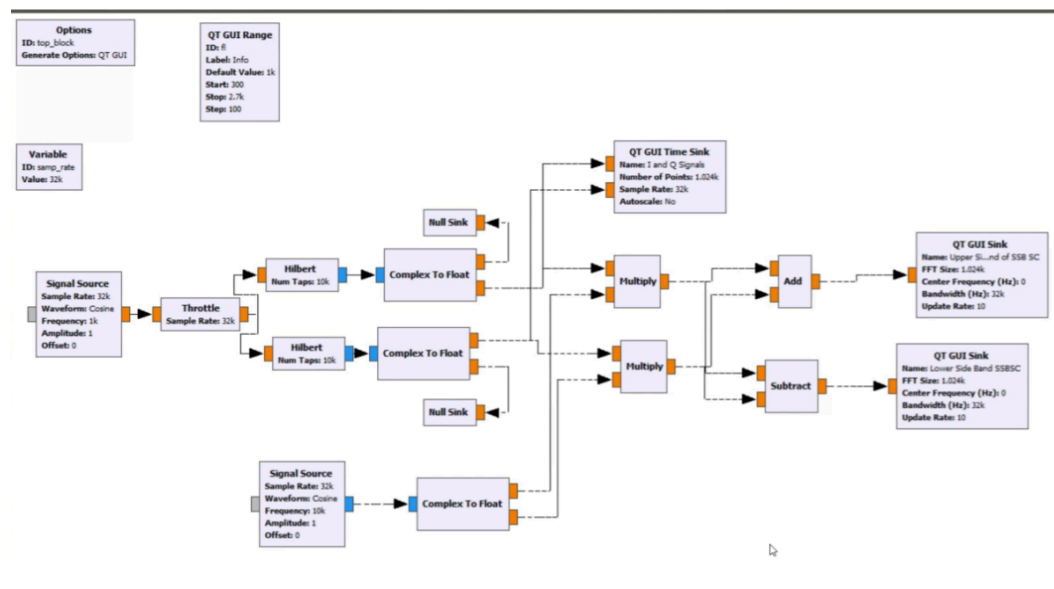
Bandwidth of SSB SC Wave

We know that the DSB SC modulated wave contains two sidebands and its bandwidth is $2f_m$. Since the SSB SC modulated wave contains only one sideband, its bandwidth is half of the bandwidth of the DSB SC modulated wave.

i.e., Bandwidth of SSB SC modulated wave = f_m

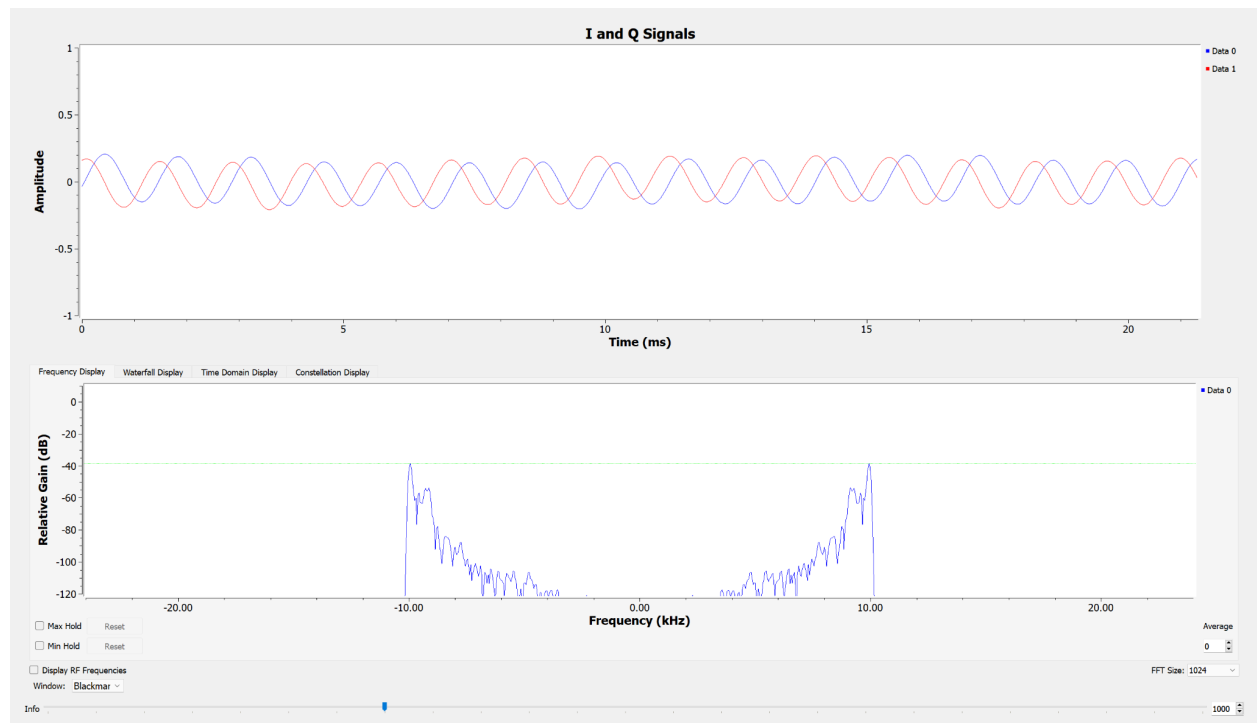
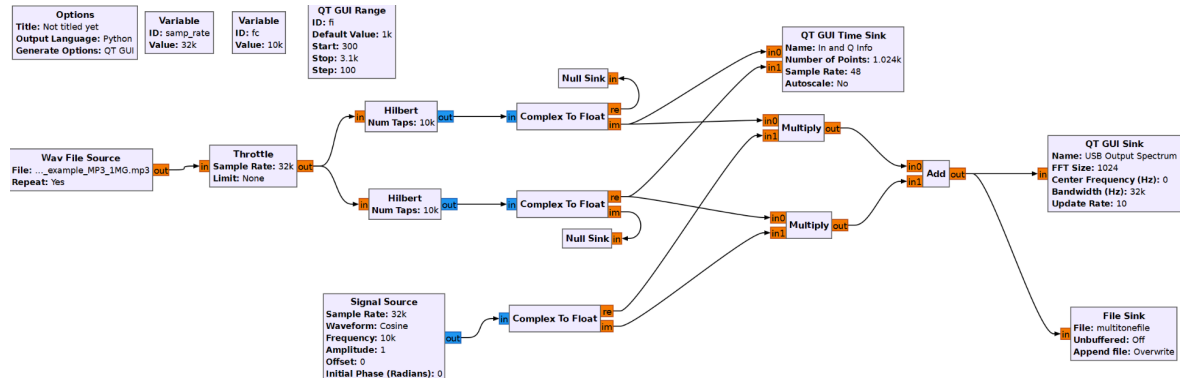
Therefore, the bandwidth of SSB SC modulated wave is f_m
 f_m and it is equal to the frequency of the modulating signal.

Single Tone Implementation:



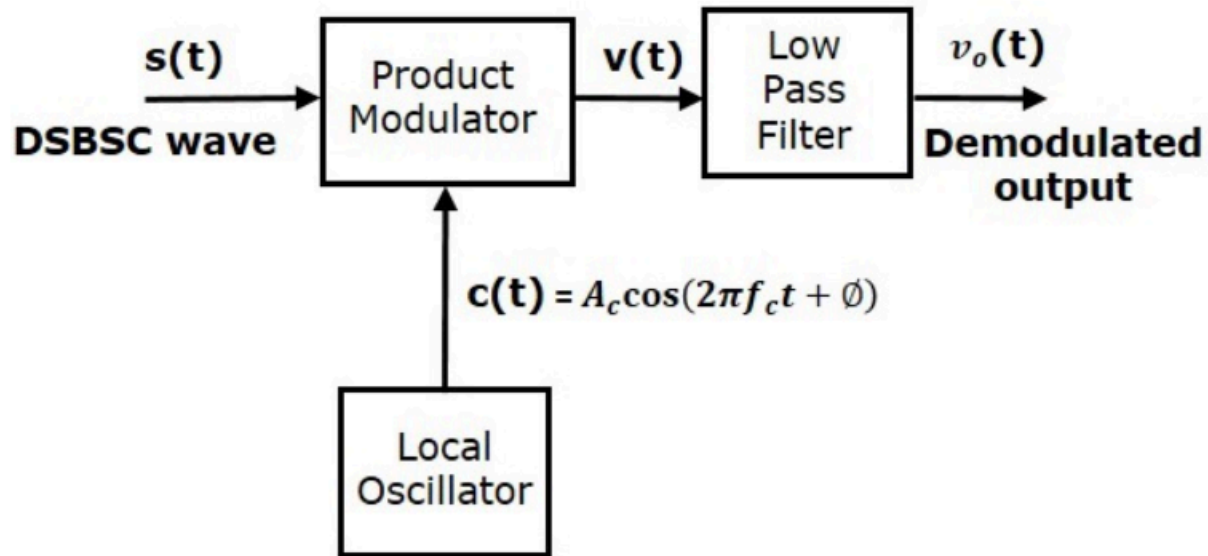
Multitone implementation:

Transmitter using Phase Discrimination Method:



Coherent Detector:

The same carrier signal (which is used for generating DSB SC signal) is used to detect the message signal. Hence, this process of detection is called as coherent or synchronous detection. Following is the block diagram of the coherent detector.



In this process, the message signal can be extracted from the DSBSC wave by multiplying it with a carrier, having the same frequency and the phase of the carrier used in DSBSC modulation. The resulting signal is then passed through a Low Pass Filter. Output of this filter is the desired message signal.

Let the DSBSC wave be

$$s(t) = A_c \cos(2\pi f_c t) m(t)$$

The output of the local oscillator is

$$c(t) = A_c \cos(2\pi f_c t + \phi)$$

Where, ϕ

ϕ is the phase difference between the local oscillator signal and the carrier signal, which is used for DSBSC modulation.

From the figure, we can write the output of product modulator as

$$v(t)=s(t)c(t)$$

Substitute,

$s(t)$ and $c(t)$ values in the above equation.

$$v(t)=A\cos(2\pi fct)m(t)A\cos(2\pi fct + \phi)$$

$$=Ac2\cos(2\pi fct)\cos(2\pi fct + \phi)m(t)$$

$$=Ac2[\cos(4\pi fct + \phi) + \cos\phi]m(t)$$

$$v(t) = Ac2[\cos(4\pi fct + \phi) + \cos\phi]m(t)$$

In the above equation, the first term is the scaled version of the message signal. It can be extracted by passing the above signal through a low pass filter.

Therefore, the output of low pass filter is

$$vot = Ac2\cos\phi m(t)$$

The demodulated signal amplitude will be maximum, when $\phi=0$.

BLOCKS USED IN THE IMPLEMENTATION:



Convert a stream of complex numbers (gr_complex) to one or two float streams. If there's only one output, it will give the real part of the complex numbers. If there are two outputs, the first will be the real part, and the second will be the imaginary part.

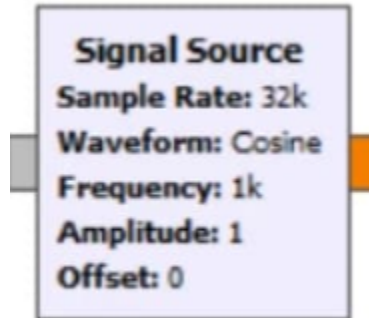


Hilbert transformer:

The real output is the input with the correct delay, while the imaginary output is the input after being Hilbert filtered (with a 90-degree phase shift).



Use as a termination point when a sink is required but we don't want to do anything with the stream.



Signal generator: produces different types of waveforms.

Sample Rate (R): Sample rate (f_s) is the average number of samples obtained in one second. Its units are samples per second or hertz e.g. 32,000 sample rate is 48 kHz.

Waveform (R)

options: Constant, Sine, Cosine, Square, Triangle, Saw Tooth

Frequency (R)

Frequency of the waveform

Amplitude (R)

Amplitude of the output

Offset (R)

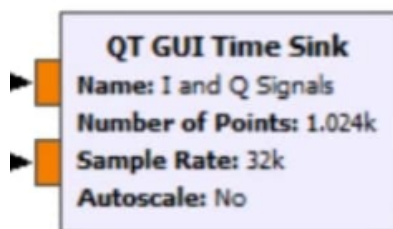
Offset from zero

Offset from zero

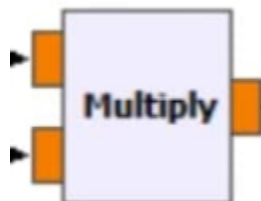
Default: 0



The throttle block either passes the input items to the output or consumes the input samples at a specified average rate, without altering the digital signal in any way.



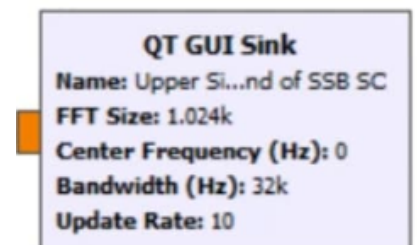
This is a QT-based graphical sink that displays sets of float or complex streams in the time domain. Each signal is shown in a different color, and the block's settings allow you to customize the label and color for each input.

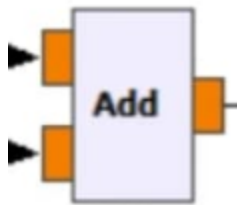


Multiply across all input streams

Parameters:

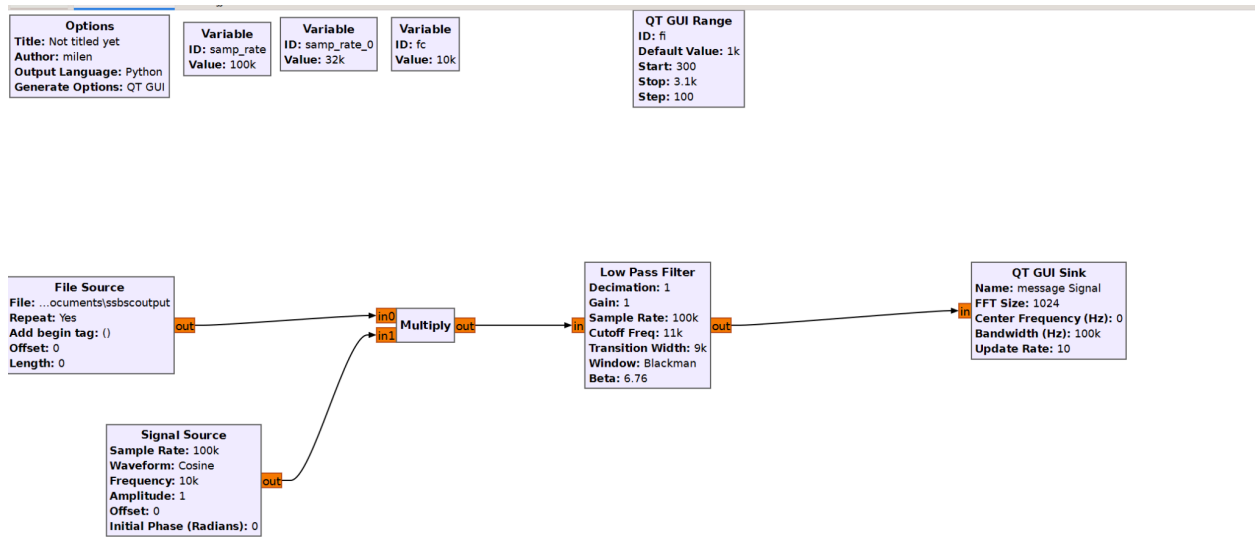
Number of streams to include in multiplication, see above.





Add samples across all input streams.

Receiver In GNURadio using Coherent Detector:



Single Tone Implementation :

